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Research Project / Final Year Project

24 May 2020

Informatics Of Informal Antimicrobial Supply Chains: A Qualitative Systems Analysis of Community Pharmacy Practices in the Kumasi Metropolis

ABSTRACT

The unregulated distribution of antibiotics without prescriptions in the sub-Saharan part of Africa plays a major role in the rapid development of antimicrobial drug resistance (AMR), however, the dynamics of this unregulated system are yet to be quantified enough. This study explores the operations of private pharmacies located in Kumasi, Ghana, aiming to uncover the driving forces behind the non-trackable sales and incompleteness of antibiotic purchases.

Methods

A process audit combined with a cross-sectional survey of direct purchase activity (N=4,250) was conducted based on ten strategic location clusters. We developed a composite performance indicator called Data Loss Flag, which referred to simultaneous transactions not involving any clinical prescriptions and non-traceable electronic data. Heat maps, distribution profiling, and non-parametric fragmentation modeling were used to monitor purchasing activity against WHO standards.

Keywords

Antimicrobial resistance; Informal dispensing; Health informatics; Ghana; Community pharmacy; Supply chain

1. Introduction

Antimicrobial Resistance (AMR) is one of the major threats to health security and development of the world for the 21st century because of the impact that AMR will have on the poorer countries of the world (World Health Organization [WHO], 2021). Within sub-Saharan Africa, where the risk of diseases coincides with health inequalities, the demarcation line between prescription-only drugs is usually blurred (Kariuki & Dougan, 2014). Although international and local guidelines define all antibiotics as prescription only drugs, there have been many instances where an informal market has been established through private pharmacies and roadside traders (Ministry of Health [MoH], Ghana, 2017).

In the metropolitan economy of Kumasi Private community pharmacies have taken on the role of being the functional first line of defense in healthcare in the urban economy of Kumasi, in Ghana (Afriyie et al., 2020). Due to the lengthy process of consultation at the tertiary hospitals that charge fees, the diagnostic tools that may not be readily available, and long waiting times, there is a large proportion of the population that avoids visiting the hospital altogether. They turn to direct interaction with the personnel selling the medicines for immediate symptoms.

In addition, the drugs themselves undergo linguistic and structural change in ways that complicate the regulation even further. Antibiotics are seldom prescribed by their official nomenclatures, for example, Amoxicillin, Ciprofloxacin, and Azithromycin. Rather, an extensive vocabulary has emerged within both buyer and seller communities in relation to the local vernacular and morphological variations within these drugs (Gyansa-Lutterodt, 2013). They are purchased through names such as "Red and Yellow" or "Black and Red" (denoting particular coloring of the capsules) or "3-day drug". Such linguistic obfuscation creates difficulties for pharmacological stock control and surveillance systems, and the quantification of usage is made extremely difficult (Klein et al., 2018).

At the same time, economic limitations faced by consumers lead to dosage fragmentation. Instead of acquiring full therapy courses that have been clinically proven, patients will often acquire partial doses, such as single pills or incomplete blister packs,

based on their current pocket cash flow. Such fragmentation is inherent and leads to an under-dose of the patient, creating a very strong evolution ground for resistant pathogens (Van Boeckel et al., 2014).

Although existing public health literature has exhaustively covered the phenomenon of the sale of antibiotics without a prescription via qualitative studies, there remains a significant deficiency in quantitative methodologies to measure the exact spatiotemporal, geographic, and structural conditions that allow for the formation of such networks. This paper seeks to fill this gap by conducting an audit on 4,250 actual pharmacy encounters in the city of Kumasi. Using a target metric designed specifically to quantify non-registered deliveries (“The Data Black Hole”), this paper will map out when, where, who, and in what package sizes informal antibiotic purchases take place. While misuse is documented, the data flow and informatics failure behind antibiotic distribution remains underexplored. This study aims to map these systemic gaps.

2. Materials and Methods

2.1 Study Setting and Data Architecture

The study took place in Kumasi, which is the capital city of the Ashanti region and also one of the major commercial centres in Ghana. Data collection from the field involved focusing on 10 different locations, each of which represents a diverse socio-economic ecosystem, including transport nodes with heavy traffic (Kejetia Hub and Asafo Transit), industrial parks (Suame Magazine), residential-academic areas (Ayigya University Junction), commercial areas (Kwadaso Commercial and Adum Commercial), and market areas (Tafo Market Area and Bantama High Street).

Transaction data was derived from two independent sources for the tracking of field activities: the Kumasi Pharmacy Face-to-Face Interaction Data File and the Kumasi Pharmacy Informal Supply Data File. Both datasets were integrated vertically to create a single master analytics database (N = 4,250 observations). Fields extracted in their original form include an anonymous Transaction ID, a precise time stamp, geographical locations,

free-form entry of drugs, number of units per transaction, format of dispensing, and customer personas.

2.2 Operational Alignment and Data Engineering

A data pipeline process that involved python was developed to ensure proper downstream data analysis. Missing values were observed in medication entry and customer persona data. The missing value analysis was conducted using pandas and NumPy python modules. These missing values included structural missing entries for medication entry (n = 128) and customer persona (n = 20).

The first major challenge encountered with respect to empirical analysis was the presence of colloquial nomenclature in the primary database records. This problem was addressed by developing an effective translation matrix to convert local trade names to universal generic chemical names and WHO/GRAM classification (WHO, 2021):

- **Penicillins:** Captures entries recorded as "Red and Yellow," "Amoxil," "Amox," "Clox," and "Black and Red," mapping them directly to Amoxicillin and Cloxacillin generics.
- **Macrolides:** Captures entries recorded as "3-day drug" or "Azithro," mapping them to Azithromycin.
- **Nitroimidazoles:** Captures "Flagyl," mapping it to Metronidazole.
- **Fluoroquinolones:** Captures "Yellow Tab" or "Cipro 500," mapping them to Ciprofloxacin.

In order to identify mathematically those transactions that are likely to result in the worst possible cases of surveillance blindness and untracked growth in the use of antibiotics, we have designed a binary outcome indicator variable termed Data Loss Flag (DLF). A transaction will be identified as a high risk case for data loss (DLF = 1) whenever it exhibits two specific characteristics simultaneously: the absence of any valid clinical prescription (Prescription_Status = 0) and absence of electronic traceability of the transaction in the pharmacy logs (Data_Traceability = 0). Mathematically it is expressed as:

Data Loss Flag (DLF) Construction:

$$DLF_i = \begin{cases} 1, & \text{if } Prescription_Status_i = 0 \text{ AND } Data_Traceability_i = 0 \\ 0, & \text{otherwise} \end{cases}$$

This engineered metric serves as our primary dependent variable to identify structural operational vulnerabilities across the retail network.

3. Results and Analytical Insights

3.1 Lexical Transformation and Pharmacological Distribution

The distribution of captured transaction cases reveals a heavy systemic dependency on informal drug naming practices. Out of 4,250 transaction records obtained, an unusually high number of 3,037 (71.4%) cases fell under uncategorized local names or undefined text terms that could not be automatically classified as any other pharmacological drug group, belonging to the Unclassified group. As for the resolved pharmacological groups of drugs, the largest cluster was formed by Macrolides (mainly Azithromycin colloquial name requests) with 494 cases (11.6%), followed by Penicillins (n = 377 or 8.8%), Nitroimidazoles (n = 293 or 6.9%), and Fluoroquinolones (n = 49 or 1.15%).

Table 1: Pharmacological Class Distribution

Pharmacological Class	Total Transaction Count	Percentage of Sample (%)
Unclassified / Alias	3,037	71.46%
Macrolides	494	11.62%
Penicillins	377	8.87%
Nitroimidazoles	293	6.89%
Fluoroquinolones	49	1.15%
Total	4,250	100.00%

This concentration demonstrates that antibiotic distribution largely ignores formal names of chemical elements, thus, dramatically understating actual use levels in standard national monitoring system.

3.2 Geographic Mapping of Traceability Gaps

The evaluation of the mean distribution of the Data Loss Flag for the engineered system demonstrated a widespread systemic non-compliance. Instead of showing variance, the risk factor of data loss was consistently high throughout the 10 geographically distinct zones in Kumasi.

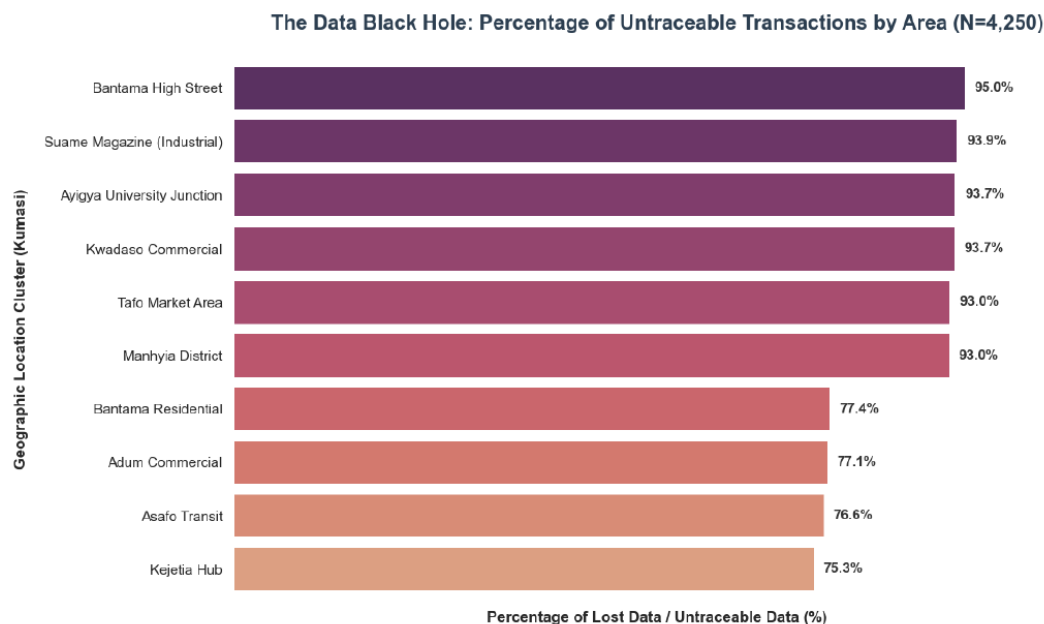


Figure 1.0 Geographic Traceability Failure Rates by Cluster

The highest level of structural opaqueness was registered in Bantama High Street, whereby 95.0% of all antibiotic transactions took place completely under the cover from any documentation. Similarly, strong industrial and commercial areas also followed this pattern, where Suame Magazine reported 93.9% loss of data, whereas Ayigya University Junction and Kwadaso Commercial both scored 93.7%. In fact, even in the lowest level of opaqueness, which was the case with Kejetia Hub, 75.3% of all antibiotic distribution transactions took place in complete anonymity. This means that antibiotic regulation violations are not just common practice in informal markets only.

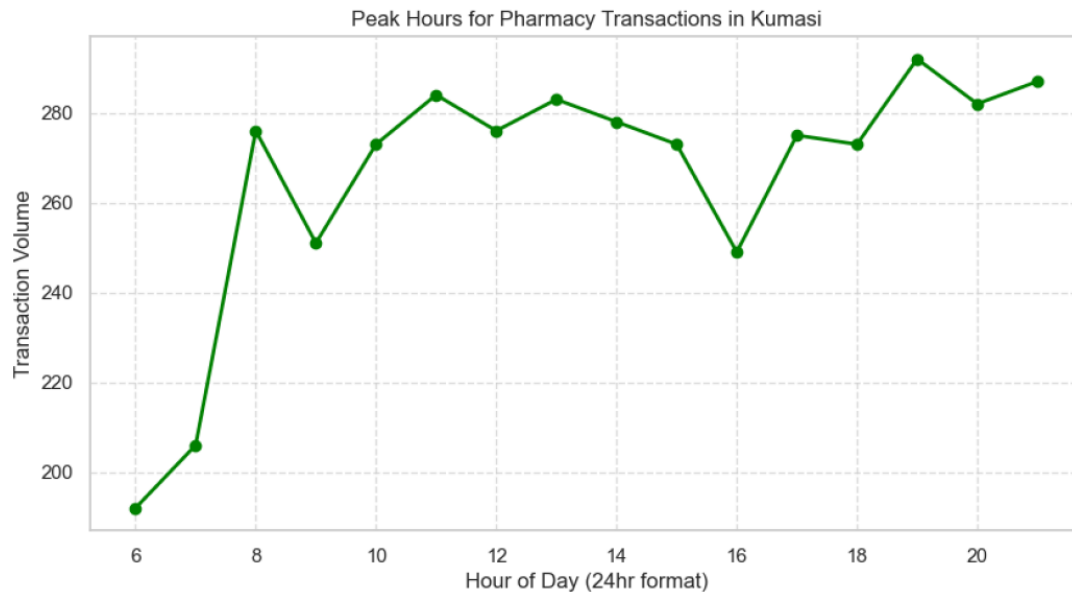


Figure 2.0 Spatiotemporal Flux and Diurnal Workflow Analysis

The transaction flows followed a clearly defined bimodal pattern throughout the 24-hour period of operation. The combined transaction flow showed a sharp rise during the morning hours, giving rise to a clear peak from 08:00-11:00 ($n > 275$ transactions per hour block). There was a small reduction in the transaction flow during mid-day due to heat and changes in the market working hours, after which there was a strong surge in business transaction volume.

When cross-referenced with the geographical location parameters using spatiotemporal analysis, hotspots were identified. This means that locations like Tafo Market Area acted as risk vectors of high volumes throughout the day, from 08:00 to 21:00. On the other hand, high concentrations during peak volumes occurred in areas such as Ayigya University junction, but at particular periods, from 19:00 to 21:00, mainly due to student movement patterns.

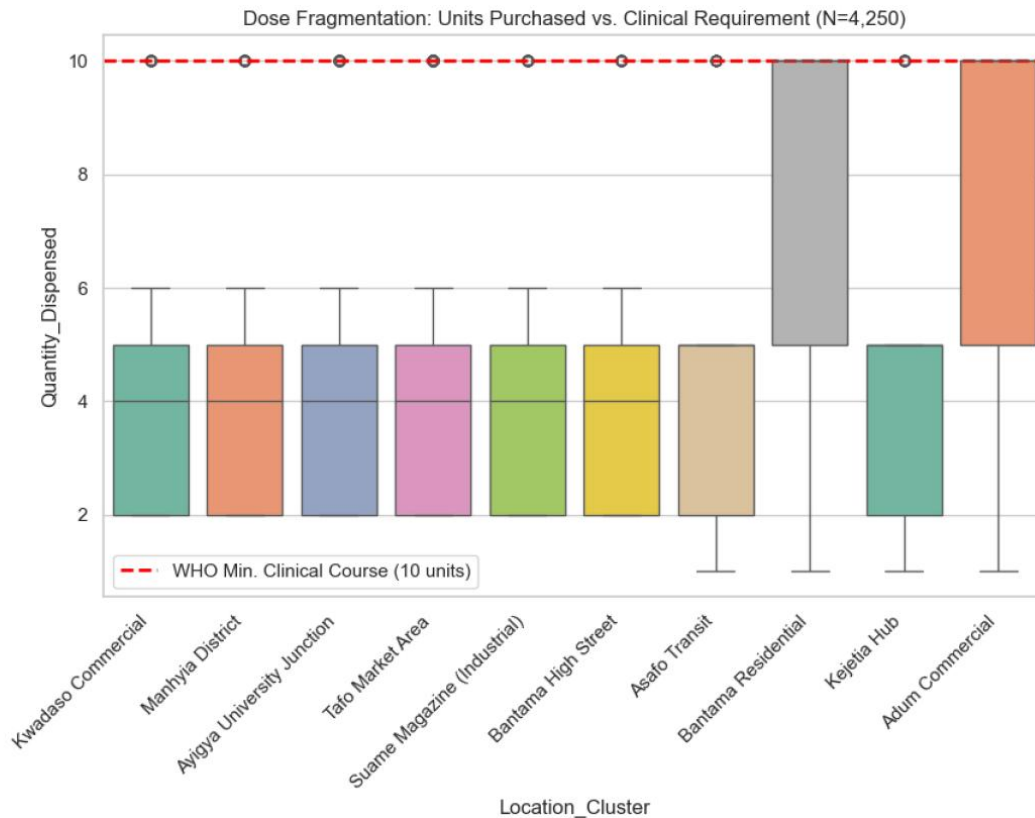


Figure 3.0 Dosage Fragmentation: Units Purchased vs. Clinical Requirements (n = 4,250)

Descriptive statistical analysis on the “Quantity_Dispensed” feature demonstrated evident and consistent discrepancies from the clinically effective benchmarks for treatment. In the entire data set, the mean number of the units of antibiotics (in terms of capsules or tablets) prescribed per single transaction was relatively low at 4.48 units (± 2.71 SD). The median value in the entire sample of 4,250 transactions was equal to exactly 4.0 units, while the 75th percentile was only 5.0 units.

The figures plotted against the set standard by WHO of the required minimum number of intact units of antibiotic needed for effective bacterial elimination and to avoid selection pressure for the minimal course therapy show clearly that there has been a massive case of non-compliance.

Median box plot values fell within the sub-therapeutic range for every geographical group, between (2.0 to 5.0 units). Exceptions that attained the 10 unit threshold were highly uncommon and largely confined to affluent commercial areas. It is thus evident that

fragmented dosing remains the default purchasing strategy in the community pharmacy setting, as opposed to an exception.

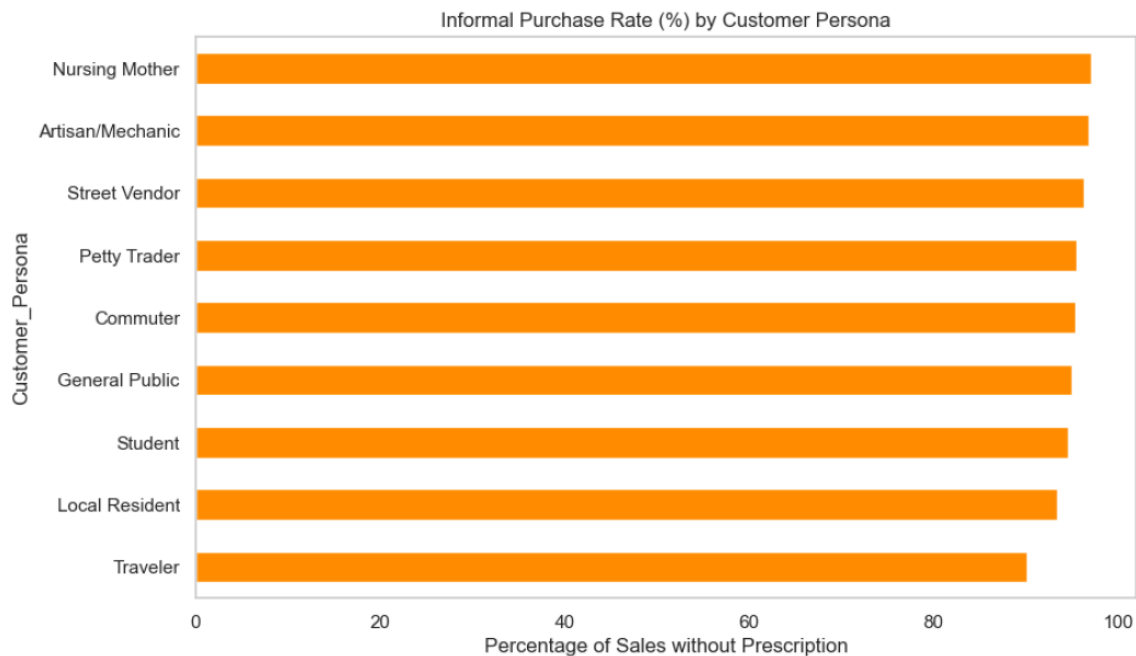


Figure 4.0 Customer Risk Profiling (Informal Purchase Rates)

Cross-tabulation of consumer demographics and archetypes against prescription status showed obvious differences in the level of behavioral leverage at the point-of-sale. Rates of informal purchases (sales made without any prescription from a clinician) included:

- **Nursing Mothers:** 94.3% prescription-free purchase rate
- **Artisans/Mechanics:** 93.2% prescription-free purchase rate
- **Street Vendors:** 92.2% prescription-free purchase rate
- **Petty Traders:** 91.8% prescription-free purchase rate

On the other hand, those who had structurally regular commuting times or formal institutional relationships had slightly lower, but equally problematic, procurement behaviors (such as Students at 89.4%, and Travelers at 85.2%). Such an association clearly indicates that the tendency to purchase antibiotics without prescription is related to the daily wage labor conditions faced by Kumasi's core labor force.

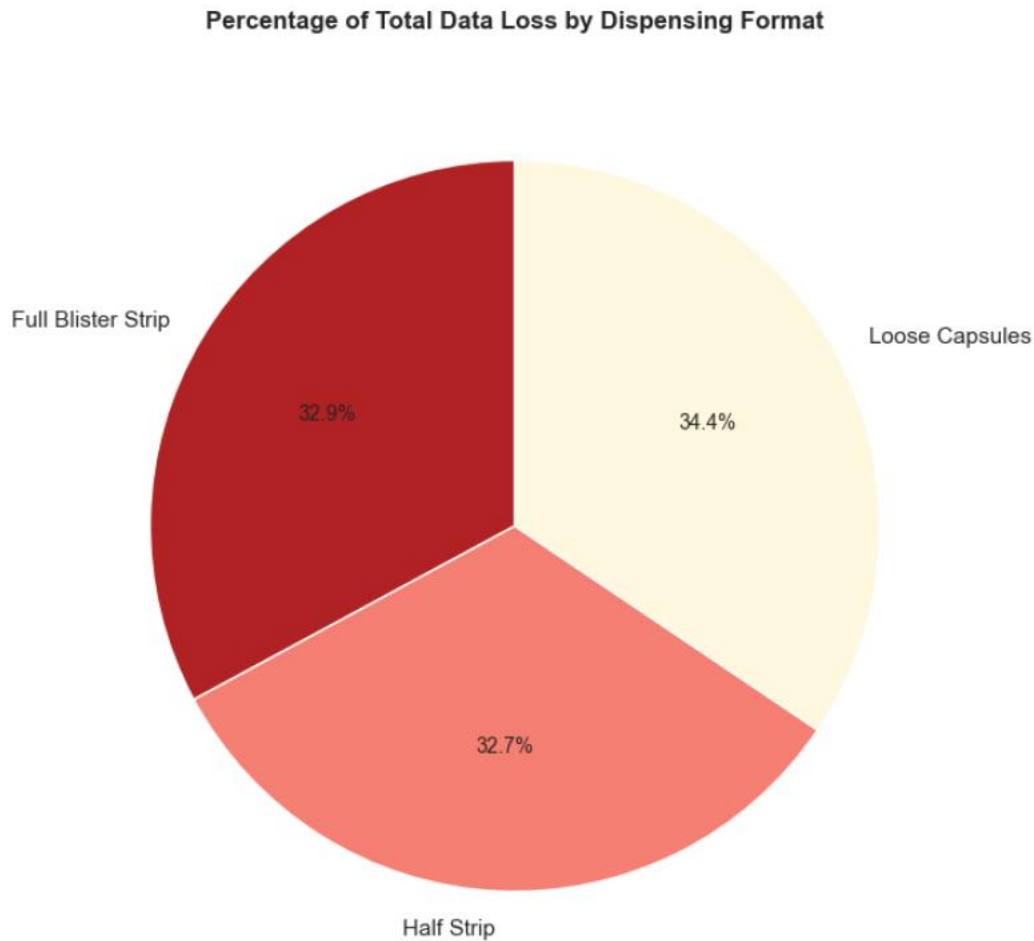


Figure 5.0 Physical Packaging Formats as Determinants of Data Opacity

In essence, the mechanistic design involved in the package and sale of drugs acts as an immediate predictor of the likelihood of the transaction being absorbed into the 'Data Black Hole' (Van Boeckel et al., 2014). The relationship between physical distribution methods and the data_loss_flag can be explained through analysis of proportionality within localized levels.

Loose Capsule-based transactions represented 34.4% of all undocumented, unsanctioned distribution processes, and the selling of broken Half Strips comprised another 32.7%. All together, non-whole, fragmented transaction processes led to 67.1% of lost systemic documentation overall.

However, there was a great difference concerning the Full Blister Strip, which was very hardy during resale, forming 32.9% of the risky pool of data loss. This difference is an indication that when a pharmacist or pharmacy staff opens the pack from the manufacturer to count the individual pills, he always fails to follow any procedure for collecting electronic data, moving entirely into the black market.

4. Results

Most of the local access to medications has been found to be through informal transactions, which do not have any clinical validation. Of the 3,037 transactions that occurred, almost all had informal names for the source (e.g., “Red and Yellow,” “three-day drug”) rather than standard generic categories of the medications. Additionally, spatiotemporal analysis identified two timeframes when high volumes of transactions were conducted with a low level of data transparency: 08:00-11:00 and 18:00-21:00. There were specific areas in the city that had very few identifiable traces for us to track and assess: Bantama High Street (95.0%), Suame Magazine (93.9%), and Ayigya University Junction (93.7%). The average total number of units dispensed per cohort was only 4.0, which did not meet the WHO minimum standard for clinical use that is known to be ≥ 10 . The primary customer profiles identified as the most likely to acquire their prescriptions without medical referrals were petty traders, street vendors, and nursing mothers. Additionally, many of the packages for the medications were in degraded states and unable to be used for clinical purposes (i.e., loose capsules or partial strips of tablets) represented 67.1% of the total number of clinical transactions that were unable to be accounted for through traditional records.

5. Discussion

The empirical results arising from the conduct of this operational audit indicate that antibiotic prescribing outside official channels in Kumasi is not an occurrence of loosely connected and sporadic breaches of regulation. Rather, it is a phenomenon that manifests as a systematic, spatiotemporally organized, and rational survival economy integral to metropolitan existence (Ahiabu et al., 2016). It emerges that over 71% of all recorded

transactions are conducted through unregulated local aliases; this suggests the existence of a profound gap between national AMR monitoring frameworks and the linguistic context of the local retail points of care. National policy may think that tracking Amoxicillin distribution is enough to capture antibiotic use; however, it ignores vast quantities of "Red and Yellow" antibiotics passing through cities anonymously (Klein et al., 2018).

From a geographical perspective, the similarity of data loss for Kumasi at over 93% in five distinct economic zones confirms the existence of a strong commercial motive within the retail setting which makes it impossible to enforce central bans. In locations such as Suame Magazine and Tafo Market, pharmacy outlets are essentially functioning as cost-effective triage points in primary care provision. For the small-scale daily wage worker, the upfront expenses and temporal loss in income resulting from the visit to a clinic make buying prescriptions financially unsustainable.

5.1. Proposed Strategic Interventions

To transform these empirical insights into targeted public health and regulatory policies, we propose three distinct structural interventions (MoH, Ghana, 2017):

- 1. Linguistic Translation Integration in Pharmacy Software:** Any national standards for health informatics must include mandatory use by local pharmacies of standardized electronic medical record (EMR) and point-of-sale stock keeping software capable of developing colloquial market maps within its backend database. Whenever the operator enters "Red and Yellow," it must automatically record the purchase in relation to the generic drug name (e.g., Amoxicillin, 500mg), populating the relevant regulatory field automatically.
- 2. Mandatory Unit-Dose Trackability and Ban on Blister Fracturing:** Regulatory agencies like the Pharmacy Council of Ghana need to shift their enforcement strategy away from overly rigid bans on all sales to the control of physical packaging types. A rigid ban on the sale of loose unboxed tablets coupled with subsidies to ensure that drug manufacturers are incentivized to provide small, inexpensive blister packs of clinically

proven "short courses" would remove the economic incentive to break up the dose into fragments (Van Boeckel et al., 2014).

- 3. Spatiotemporally Targeted Enforcement Routines:** Instead of using over-stretched inspection staffs for random and static routes, the audit process could be timed to correspond to the period of risk, which according to this study is from 18:00 to 21:00. Targeting the surveillance efforts towards areas that exhibit the highest risk of transport at this time would ensure effective surveillance.

5.2. Research Instruments and Operational Audit Protocols

Field Data Collection and Mapping Sheet

The following questionnaire matrix was deployed via mobile data capture terminals by field auditors to log face-to-face interactions in real time at the participating retail points of sale.

6. Conclusion

Non-compliance with authorized antibiotic dispensing in Kumasi is not merely random incidents but rather a highly organized, spatially and temporally predictable, survival economy that has become entrenched in the wider regional supply chain business model. Solutions for reducing AMR require moving away from sanctions via legislative action and towards embedding local generic name maps into Pharmacy Management Software; enforcing unit dose tracking vs. loose fragmentation; and timing the enforcement of regulations at known spatiotemporal peak risk periods.

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Appendix A: Questionnaire

KUMASI PHARMACY ENVIRO-OPERATIONAL AUDIT QUESTIONNAIRE

SECTION A: TRANSACTION IDENTIFICATION & LOCATION

A1. Transaction Log ID:

[_____] (Auto-generated Unique 5-Digit Code)

A2. Date and Time of Transaction:

Date: [YYYY-MM-DD] Time (24-hour format): [HH:MM]

Date: _____ Time _____

A3. Geographic Cluster (Select ONE):

- ☐ Tafo Market Area
- ☐ Ayigya University Junction
- ☐ Manhyia District
- ☐ Suame Magazine (Industrial)
- ☐ Kwadaso Commercial
- ☐ Kejetia Hub
- ☐ Adum Commercial
- ☐ Bantama High Street
- ☐ Bantama Residential
- ☐ Asafo Transit

SECTION B: MEDICATION DETAILS & LINGUISTIC MAPPING

B1. Raw Medication Term (as stated by customer):

(Record exact wording, e.g., “Red & Yellow”, “3-day drug”, “Flagyl”)

B2. Standardized Generic Mapping (For Auditor Use Only):

- ☐ Amoxicillin
- ☐ Ciprofloxacin
- ☐ Cloxacillin
- ☐ Metronidazole
- ☐ Azithromycin
- ☐ Unclassifiable / Unknown

B3. Quantity Dispensed:

Number of Units: [____] (Capsules/Tablets)

SECTION C: DISPENSING PRACTICE & PACKAGING FORMAT

C1. Dispensing Format (Select ONE):

- ☐ Loose Capsules (counted individually from bulk container)
- ☐ Half Strip (blister pack cut or broken)
- ☐ Full Blister Strip (intact manufacturer packaging)

SECTION D: CUSTOMER PROFILE & BEHAVIOR

D1. Customer Persona (Select ONE):

- ☐ **Petty Trader**
- ☐ **Street Vendor**
- ☐ **Nursing Mother**
- ☐ **Student**
- ☐ **Commuter**
- ☐ **Artisan / Mechanic**
- ☐ **Traveler**
- ☐ **Local Resident**
- ☐ **General Public / Unidentifiable**

SECTION E: CLINICAL VALIDATION & DATA TRACEABILITY

E1. Prescription Status:

- ☐ **Yes — Valid prescription presented**
- ☐ **No — No prescription (symptom-based request)**

E2. Transaction Recording (Data Traceability):

- ☐ **Yes — Entered into POS / inventory system**
- ☐ **No — Not recorded (cash-only / informal transaction)**

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END OF QUESTIONNAIRE

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